Dynamic Multi-Channel Random Access Procedure for 6G-Enabled Ambient IoT

Shilun Song, *Member, IEEE*, Huan Zhang, Huiyang Xie, and Hu Jin, *Senior Member, IEEE*

***Abstract*—This paper presents a comprehensive study on random access (RA) design for multi-channel Ambient IoT (A-IoT) networks enabled by uplink frequency division multiple access (FDMA). We introduce and unify the three-step RA procedure (RAP) specified in ongoing 3GPP standardization, which serves as a useful reference for future research. To fully exploit the multi-channel capability, we propose enhanced frame structures, including a novel multi-round Msg1 collection (MRMC) scheme, in which Msg1 carries a randomly generated ID from each device and is transmitted in the first step of the three-step RAP. We also develop new analytical models that capture the unique characteristics of multi-channel A-IoT, providing closed-form expressions for RA efficiency that serve as a foundation for network optimization. Furthermore, we design online control algorithms that estimate the device population in real time and dynamically adjust frame configurations for optimal operation under device uncertainty. The proposed Boosted Estimation with Early Frame Termination (BE^{2}FT) algorithm effectively handles bursty traffic conditions and improves resource utilization. Extensive simulations validate the accuracy of our analytical models and demonstrate significant gains in RA efficiency and service time. The proposed scheme achieves performance close to the ideal performance, where the number of devices is assumed perfectly known and the frame length is optimally configured.**

***Index Terms*—Ambient IoT, random access, FDMA, backscatter communication.**

I. Introduction

OVER the past decade, the Internet of Things (IoT) has evolved from a concept to a central element in global communication infrastructure. The widespread deployment of IoT devices, from smart homes to industrial systems, has redefined connectivity. As the number of connected devices continues to grow, ensuring low power consumption, wide-area coverage, and massive access has become a significant challenge [1]. To meet these demands, the 3rd Generation Partnership Project (3GPP) introduced three major IoT categories: Broadband IoT for high-rate, low-latency applications; Critical IoT for ultra-reliable services; and Massive IoT for large-scale, low-cost, low-power devices [1]. However, all still depend on batteries, and battery replacement incurs considerable maintenance costs. This raises a key question: Can ambient energy be harnessed to support communication?

Two main approaches have emerged: energy harvesting and radio frequency (RF) backscatter. Energy harvesting enables devices to operate independently by drawing energy from ambient sources, such as solar, RF, thermal, or mechanical energy, utilizing transducers and power management modules [2]. RF backscatter, on the other hand, enables ultra-low-power communication by modulating and reflecting existing RF signals through changes in load impedance, thereby avoiding the need for active components [3]. While often grouped under energy harvesting, backscatter is distinct in that it utilizes ambient RF not only as a power source but also as a carrier.

To extend the use of ambient energy, 3GPP initiated the Ambient IoT (A-IoT) project as a key feature of 5G Advanced and future 6G networks [4]. The initial study was completed in late 2024 [5], with specification work starting in 2025 [6]. A-IoT devices are mainly categorized into three types based on their power and transmission capabilities [5], [7]: Type 1 devices are fully passive and rely on RF energy for backscatter communication, operating at around one microwatt peak power, while Type 2b and Type C devices support energy storage and active transmission using an internal carrier, with peak power consumption of up to several hundred microwatts for Type 2b and up to a few milliwatts for Type C. These types correspond to different application scenarios and traffic characteristics. Type 1 devices are typically used for large-scale inventory and identification applications, where the reader, an active transceiver that emits a continuous wave (CW) and decodes the backscattered responses from devices, triggers simultaneous responses from numerous devices, resulting in inherently bursty traffic. Type 2b and C devices, on the other hand, target sensor-oriented or event-driven applications, where data are generated locally and transmitted independently. Current 3GPP standardization efforts focus on Type 1 devices [6], whereas discussions on Type 2b and Type C operations are planned for future 3GPP releases. Equipped with minimal energy storage and simple envelope detectors, Type 1 devices transmit by passively backscattering external carriers, similar to radio-frequency identification (RFID) tags used for identification and tracking. This paper also focuses on Type 1 A-IoT systems.

In 6G, IoT systems are expected to meet more stringent requirements, including higher peak data rates, lower air-interface latency, enhanced reliability, and support for massive

Received 20 June 2025; revised 4 December 2025; accepted 7 December 2025. Date of publication 12 December 2025; date of current version 20 February 2026. This work was supported by the Brain Pool Program funded by the Ministry of Science and ICT through the National Research Foundation of Korea under Grant RS-2025-25456394. *(Corresponding authors: Huiyang Xie; Hu Jin.)*

The authors are with the Department of Electrical and Electronic Engineering, Hanyang University, Ansan 15588, South Korea (e-mail: shilunsong@hanyang.ac.kr; cola1999@hanyang.ac.kr; xiehuiyang@hanyang.ac.kr; hjin@hanyang.ac.kr).

Digital Object Identifier 10.1109/JSAC.2025.3643814

1558-0008 © 2025 IEEE. All rights reserved, including rights for text and data mining, and training of artificial intelligence and similar technologies. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Higher School of Economics. Downloaded on April 07,2026 at 10:42:35 UTC from IEEE Xplore. Restrictions apply.

connectivity [8]. Within this context, A-IoT serves as a specialized solution that prioritizes ultra-low cost and minimal device complexity while enabling extremely high connection densities of up to 10^7 devices per square kilometer or more, with Type 1 devices representing its primary implementation. With passive backscatter operation and reader-initiated communication, A-IoT is well suited for large-scale, low-power applications such as warehousing, item-level tracking, and inventory management [9]. In such scenarios, traffic is inherently bursty because a large number of devices may be activated simultaneously upon the reader's triggering, and the device identities are initially unknown, making scheduled access impractical. These characteristics render random access (RA) one of the most critical capabilities for A-IoT.

In existing ultra-high frequency (UHF) RFID systems, RA typically follows a three-step random access procedure (RAP) [10]: 1) the tag sends a 16-bit random ID (RID); 2) the reader echoes the RID if successfully decoded; and 3) the tag replies with its unique identification (UID). All steps occur on a single channel. While 3GPP adopts a similar RAP for A-IoT [6], it has officially incorporated uplink (UL) frequency division multiple access (FDMA) into the standardization framework, enabling simultaneous responses from multiple devices and significantly improving scalability compared to legacy RFID. 3GPP refers to the RID response in the first step and UID response in the third step as Msg1 and Msg3, respectively. The downlink (DL) message in the second step, termed Msg2, may carry one or more RIDs to facilitate the Msg3 responses through UL FDMA.

Despite its potential to enhance scalability via simultaneous multi-device responses, the RA design for UL FDMA-enabled multi-channel A-IoT networks remains largely unexplored. Several key challenges arise in this context. *First*, how should the frame and slot structures in the three-step RAP be designed to fully leverage the benefits of the multi-channel property? *Second*, how should contention be managed during multi-channel RA in the three-step RAP? *Third*, how can UL multi-channel resources be efficiently utilized under uncertainty in the device population? These questions have not been systematically addressed within the current A-IoT framework. This work aims to fill this gap through the design of protocols, analytical models, and access algorithms.

Through addressing these challenges, this paper makes the following key contributions:

1) *Introduction of the recent A-IoT RAP in 3GPP:* This work is the first to introduce the 3GPP A-IoT three-step RAP in the literature. In particular, we integrate scattered details from the 3GPP standardization process into a clear and unified description to facilitate understanding and serve as a reference for future A-IoT research.
2) *Enhanced frame structure for multi-channel A-IoT:* In addition to the frame structures specified in 3GPP, we propose an enhanced frame structure incorporating a novel multi-round Msg1 collection scheme to further exploit the UL FDMA-enabled multi-channel capability of A-IoT and improve system performance.
3) *Analytical models:* We develop new mathematical models to analyze the RA efficiency of A-IoT networks, explicitly addressing the multi-channel capability. This capability makes existing analytical models for conventional RFID systems inapplicable, necessitating new formulations. The proposed models serve as the foundation for designing algorithms that enable network optimal operation.
4) *Online control accommodating device population uncertainty:* We propose online control algorithms that dynamically configure the frame length without requiring prior knowledge of the number of devices. The algorithms estimate the number of remaining devices contending for the channel in real time and adapt the frame length accordingly to achieve optimal operation.
5) *Extensive simulations:* We conduct extensive simulations to show the accuracy of the proposed analytical models and the effectiveness of the proposed online control algorithms.

The rest of the paper is organized as follows. Section II reviews existing RA technologies related to the 3GPP A-IoT RAP. Section III introduces the system model of multi-channel A-IoT and the associated RAP schemes. Section IV presents a detailed performance analysis of the RAPs. Section V describes the design of the estimation methods and proposes the online control algorithms. Section VI evaluates the performance of the proposed RAP schemes with online control algorithms through extensive simulations. Finally, Section VII concludes the paper.

II. Related Work

Recently, 3GPP has started to specify the basic signaling and procedures for the three-step RAP in A-IoT. In parallel, existing literature on A-IoT has primarily focused on interpreting the potential and challenges of the 3GPP standards [1], [4] and exploring application requirements in practical scenarios [11], [12]. To date, only a few studies have addressed A-IoT, and these have mainly concentrated on physical layer issues, such as channel and signal design [13] and coverage enhancement [14]. 3GPP's work item identifies the design of a complete RAP in the medium access control (MAC) layer as a key objective, which remains under consideration in ongoing releases [6]. To the best of our knowledge, no prior study has proposed or analyzed the three-step RAP for multi-channel A-IoT. This work is the first to fill that gap by introducing the three-step RAP, enhancing the frame structure, analyzing its performance, and designing algorithms for optimal A-IoT operation.

A-IoT comprises multiple device types with distinct power and transmission capabilities: Type 1 devices operate through passive backscatter with reader-initiated communication, while Type 2b and C support active transmissions and can independently generate data. These differences lead to two representative RA models: 1) one triggered by the reader and 2) the other driven by device-originated activity. Accordingly, this section reviews existing studies in two main directions. The first part focuses on RA mechanisms for *reader-initiated A-IoT*, which is the primary focus of this work, with an emphasis on framed slotted ALOHA optimization. The second part surveys RA schemes for *device-initiated A-IoT*, which,

Authorized licensed use limited to: Higher School of Economics. Downloaded on April 07,2026 at 10:42:35 UTC from IEEE Xplore. Restrictions apply.

although not the main focus of this paper, provide valuable insights for the broader design of RA protocols in A-IoT. It is also worth noting that RAPs for Type 2b and C devices have not yet been discussed in the 3GPP standardization process, and consequently, related academic studies are still limited.

A. RA for Reader-Initiated A-IoT

Before 3GPP's proposal on Type 1 A-IoT, reader-initiated communication was mainly employed in RFID systems [10]. Consequently, insights from legacy RFID RA schemes can provide valuable guidance for designing effective RA mechanisms for A-IoT, as they are consistent with 3GPP's approach to A-IoT standardization. While system scale and channel configurations differ, techniques such as frame-based random access, backlog estimation, and slot scheduling remain relevant for A-IoT RA optimization and offer a strong conceptual foundation for A-IoT RA design.

Among RFID RA control schemes, dynamic frame slotted ALOHA (DFSA) is the most widely used, and current UHF RFID protocols natively support it [10]. DFSA operates based on the number of slots per frame broadcast in the reader's Query message, with tags randomly selecting slots [15]. Early DFSA studies, resembling classic Slotted ALOHA, assumed constant slot lengths [15], [16]. With UHF RFID, slot durations began to vary depending on transmission outcomes [17]. Research has consistently focused on two key aspects: determining the optimal number of slots per frame and estimating the number of devices. Maximizing throughput requires the reader to select an appropriate number of slots per frame while considering variations in slot duration and their associated probabilities [18].

Estimating the number of unserved tags has long been a core challenge. Since the number of tags is initially unknown, readers must infer it from channel outcomes. Schoute [19] and Vogt [15] proposed low-complexity estimation algorithms that multiply collision counts by a constant of 2.39 and 2, respectively. Later, Vogt introduced a Chebyshev-based estimation algorithm that uses the numbers of idle, success, and collision slots to estimate a distribution and select the most likely number of tags [20]. Although more accurate, this method is computationally intensive for large or unbounded populations. As a result, Schoute's simpler algorithms remain common in recent works [18]. The early break mechanism [17], [21] further improved DFSA's flexibility and stability by allowing the reader to interrupt the RA process within a frame if it detects the frame is too long or too short.

All the above discussions consider systems with single-channel operation. In a recent development, Huang et al. proposed an FDMA extension to standard UHF RFID, enabling simultaneous multi-tag inventory [22]. By assigning different Miller subcarrier frequencies and applying successive interference cancellation (SIC) at the reader, their design achieves higher throughput while maintaining compatibility with current RFID standards. This work aligns closely with the multi-channel A-IoT vision, demonstrating that FDMA can significantly enhance identification performance. However, RA schemes that could further improve efficiency were not considered in their design.

B. RA for Device-Initiated A-IoT

As 3GPP plans to discuss the RAP for Type 2b and C devices in future releases, only limited details about their operations are currently available in the standards. Nevertheless, based on the characteristics of A-IoT, two aspects are considered particularly important: 1) energy harvesting, where a device initially remains in a sleep state with only its harvesting circuitry active and switches to an active state to listen and transmit after sufficient energy has been accumulated [5]; and 2) device-initiated transmission, which requires the reader to frequently broadcast RA configuration information so that newly awakened devices can perform UL transmissions [7].

A substantial body of work on RA in energy-harvesting IoT networks already exists in the literature. However, these studies are not specifically tailored to the RAP design for these devices because the corresponding 3GPP specifications have not yet been established. First, the slotted ALOHA protocol, as one of the most fundamental RA schemes, was investigated in [23], where devices transmit only after harvesting sufficient energy from the base station (BS). If uplink activity is detected at the beginning of a slot, the BS continues to listen; otherwise, it uses the remaining portion of the slot to deliver energy to the devices. Later, coded ALOHA schemes with collision resolution were introduced. In particular, irregular repetition slotted ALOHA (IRSA) transmits the same packet multiple times [24], whereas coded slotted ALOHA (CSA) divides packets into segments and encodes them across time slots using short codes [25]. Considering the energy harvesting rate and battery capacity, IRSA and CSA have been studied for throughput analysis and parameter optimization [26], [27].

Although device-initiated transmissions in 5G cellular IoT systems differ from those in A-IoT, related studies provide valuable insights for the future design of device-initiated A-IoT. The four-step RAP for 5G IoT systems builds upon the concept of multi-channel slotted ALOHA, where the access class barring (ACB) mechanism controls the transmission probability. The optimal online control of the ACB factor to maximize access efficiency in massive IoT scenarios was introduced in [28], achieving a per-channel throughput of 0.368 packets/slot. Alternatively, machine learning-based algorithms have also been proposed to achieve comparable performance [29]. For the coexistence of heterogeneous IoT applications, ACB-based prioritization methods were investigated in [30], while [31] explored energy-efficient ACB mechanisms. In addition, [32] examined ACB optimization for energy-harvesting IoT networks. Subsequent works introduced the splitting concept into the four-step RAP, further improving access efficiency with a per-channel throughput of up to 0.468 packets/slot [33]. Later studies also investigated the joint optimization of the ACB mechanism and backoff procedure [34]. While the above schemes provide valuable insights, their RA structures are not based on framed random access. Therefore, their concepts cannot be readily applied or meaningfully compared to the reader-initiated A-IoT scenario.

III. System Model

In Section III-A, we first introduce the channel structure of A-IoT, highlighting the FDMA capability that fundamentally

Authorized licensed use limited to: Higher School of Economics. Downloaded on April 07,2026 at 10:42:35 UTC from IEEE Xplore. Restrictions apply.

TABLE I
SUPPORTED COMBINATION OF BIT DURATION T_b AND CORRESPONDING SET OF AVAILABLE R VALUES

Bit duration T_b (μs)	The set of available R values
266.67	{1, 2, 4, 8, 16, 32, 64, 128}
133.33	{1, 2, 4, 8, 16, 32, 64}
66.67	{1, 2, 4, 8, 16, 32}
33.33	{1, 2, 4, 8, 16}
16.67	{1, 2, 4, 8}
8.33	{1, 2, 4}
4.17	{1, 2}
1.39	{1}

distinguishes it from conventional RFID and necessitates corresponding adjustments in RAP. Then, we describe the basic RAP in Section III-B, followed by its extended version in Section III-C, which aims to better exploit FDMA's potential advantages. Finally, to fully maximize the benefits provided by FDMA, we propose a multi-round Msg1 collection (MRMC) scheme in Section III-D.

A. Uplink FDMA in A-IoT

In A-IoT access networks, two physical channels are introduced to facilitate network operations: the Physical Reader-to-Device Channel (PRDCH) and the Physical Device-to-Reader Channel (PDRCH), commonly referred to as the DL and UL channels, respectively [5]. These two channels are separated in frequency by employing frequency-division duplexing (FDD). The reader transmits control signals over the DL channel, while continuously emitting a CW in the UL channel to enable the devices' backscatter communication. This CW simultaneously serves as an energy source and as a carrier wave for devices' responses. Moreover, the UL channel employs FDMA to support concurrent responses from multiple devices.

To enable simultaneous responses from multiple devices, 3GPP introduces an uplink FDMA mechanism in A-IoT, representing a significant departure from the single-channel architecture of legacy RFID systems. This mechanism utilizes small frequency shift (SFS), implemented via line code repetition [35]. During initialization, the reader broadcasts two key parameters: the bit duration T_b and a set of allowable repetition factors R, as shown in Table I. Each device encodes its data using Manchester coding and repeats each encoded bit sequence R times within duration T_b, resulting in a periodic square-like waveform. The frequency band occupied by this waveform depends on the selected R, thus allowing devices to respond concurrently in distinct spectral regions.3GPP discussed the feasibility of small frequency shift (SFS)-based FDMA in TR 38.769 [5]. As described in TR 38.769, although non-ideal device clocks may introduce sampling frequency offset (SFO), potentially causing partial overlap between adjacent sub-channels, the resulting interference can be mitigated through proper guard-band sizing and SFO-aware frequency assignment.

This repetition-based modulation enables FDMA-style responses without requiring explicit frequency synthesizers. Each R value implicitly defines a logical sub-channel, allowing multiple devices to concurrently respond in separate spectral regions. However, the number of usable R values, directly corresponding to the number of simultaneous device responses, is inherently limited. Spectral overlap among harmonics, sampling rate constraints, and backscatter circuit resolution collectively restrict the achievable frequency-domain separation. As a result, current 3GPP evaluations indicate that up to eight sub-channels can be reliably supported without significant inter-channel interference [35].

This new capability, enabling simultaneous UL responses from multiple devices, represents a major advancement beyond conventional RFID systems and fundamentally reshapes RAP design requirements. Accordingly, future RAP schemes should fully leverage this FDMA-enabled parallel response capability to maximize throughput and efficiency in large-scale A-IoT deployments.

B. Basic Random Access Procedure (B-RAP) in A-IoT

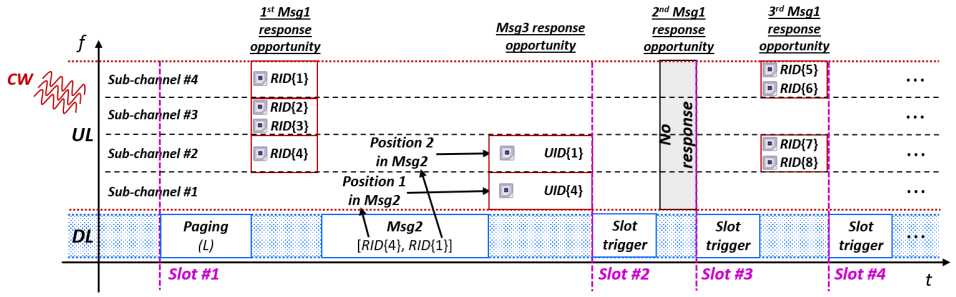
Consider an A-IoT system supporting up to K simultaneous device responses in the UL, where $K \in [1, 8]$ denotes the number of FDMA-enabled sub-channels.

Fig. 1(a) illustrates a basic frame structure of an A-IoT system. A frame consists of multiple consecutive slots, and the total number of slots is determined by the parameter L contained in the *paging message* transmitted in the DL. The parameter L specifies the number of *Msg1 response opportunities* available within a frame. As shown in Fig. 1(a), each Msg1 response opportunity has K sub-channels, and each sub-channel can be independently accessed by devices to transmit their Msg1 signals. Therefore, increasing L provides more Msg1 transmission opportunities within a frame, enabling more devices to attempt RA. In the basic RAP (B-RAP), since each slot contains one Msg1 response opportunity, the number of slots equals L. Besides, the paging message also conveys the value of K. Each subsequent slot within the frame is individually triggered by a shorter *slot trigger* message from the reader.

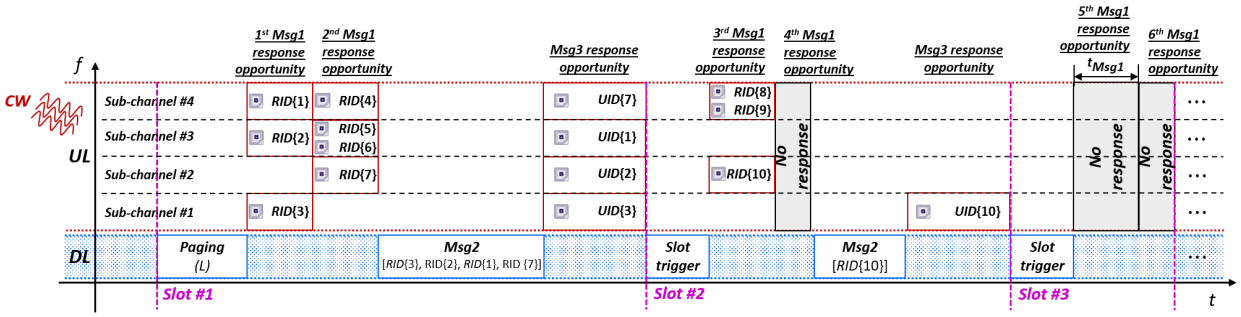
Upon receiving the paging message containing the parameter L, all triggered devices become active and perform the following procedures in the same frame: Each device uniformly and randomly selects an integer from the set $\{0, \ldots, L-1\}$ and stores it as its *backoff counter*. This simultaneous activation results in bursty traffic, while the reader, without prior knowledge of the number of triggered devices, faces additional challenges in operating the RAP. If the backoff counter is initially zero, the device immediately responds with Msg1 containing its RID on one of the K available sub-channels in the Msg1 response opportunity. Otherwise, the device decrements its backoff counter by one upon receiving each subsequent slot trigger and responds with Msg1 when the counter reaches zero. Consequently, each device makes exactly one Msg1 response per frame; if this response fails, the device waits until the next frame.

Due to the random selection of slots and sub-channels, Msg1 responses within each slot may collide, potentially causing the reader to fail in decoding multiple RIDs.After receiving the Msg1 responses, the reader identifies the successfully decoded RIDs. These RIDs are included in Msg2,

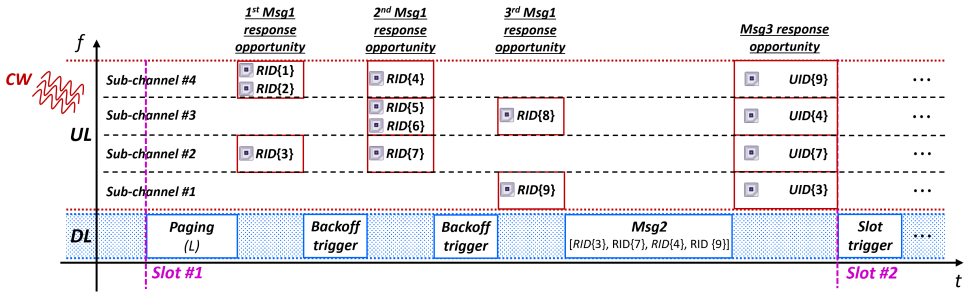
Authorized licensed use limited to: Higher School of Economics. Downloaded on April 07,2026 at 10:42:35 UTC from IEEE Xplore. Restrictions apply.



(a) Basic random access procedure (B-RAP) of A-IoT when $K = 4$.



(b) Extended random access procedure (E-RAP) of A-IoT when $K = 4$.



(c) Proposed multi-round Msg1 collection random access procedure (MRMC-RAP) of A-IoT when $K = 4$.

Fig. 1. Random access procedure in A-IoT.

which is broadcast by the reader. The ordering of RIDs in Msg2 specifies the sub-channel indices to be used for the subsequent Msg3 responses. The devices acknowledged in Msg2 then transmit Msg3, carrying their UID, on the assigned sub-channels and complete the current slot, as illustrated in Slot 1 of Fig. 1(a).The time-domain resource to respond to a Msg3 is defined as *Msg3 response opportunity*. Consequently, contention occurs only during Msg1 responses, while Msg3 responses are fully scheduled and collision-free. Note that due to FDMA support, the maximum number of successful Msg3 responses per slot is K.

If the reader detects no device responses at all, as illustrated in Slot 2 of Fig. 1(a), it quickly identifies this absence since the detection duration is shorter than a full Msg1 response. Similarly, if responses occur but no valid RID is decoded due to collisions on the selected sub-channels (as shown in Slot 3), the reader also treats this as a failed Msg1 response. In both scenarios, the reader promptly initiates the next slot by transmitting the subsequent slot trigger message.

This structured process repeats until all L slots within the frame have been completed. Then, the reader can send a new paging message to indicate a new frame for the remaining devices, where the parameter L can differ from the last one.

C. Extended Random Access Procedure (E-RAP) in A-IoT

One limitation of the random access procedure introduced above is that it does not fully exploit the capability for simultaneous device responses provided by uplink FDMA. Specifically, due to random sub-channel selection during Msg1 responses, collisions frequently occur, substantially reducing the number of successful responses compared to the available K sub-channels. When the number of devices is large, the average maximum number of successfully decoded Msg1 responses is approximately $e^{-1}K \approx 0.368\ K$ [28].

Consequently, when devices subsequently respond with Msg3, on average only $0.368\ K$ sub-channels are effectively utilized, leaving the remaining $(1 - e^{-1})K \approx 0.632K$ sub-channels unused. This underutilization represents significant inefficiency, especially considering the relatively long duration of Msg3 responses. To improve resource utilization, 3GPP introduces an extended RAP (E-RAP) featuring two consecutive Msg1 response opportunities per slot ($X = 2$), as

Authorized licensed use limited to: Higher School of Economics. Downloaded on April 07,2026 at 10:42:35 UTC from IEEE Xplore. Restrictions apply.

illustrated in Slot 1 of Fig. 1(b). The value of $X = 2$ is indicated in the paging message to instruct device operation, while $X = 1$ corresponds to the B-RAP described in Section III-B. With $X = 2$, the average number of successful Msg1 responses approximately doubles to $2e^{-1}K \approx 0.736\ K$, significantly enhancing sub-channel utilization during Msg3 responses. This capability enabled by FDMA represents a notable advancement compared to conventional RFID.

Since there are L Msg1 response opportunities in the frame initiated by the paging message, and each slot contains two of them, the number of slots in the frame is $\lceil \frac{L}{2} \rceil$. Upon receiving the paging message with L, each device uniformly and randomly selects an integer from the set $\{0, \ldots, L-1\}$ and stores it as its backoff counter. The difference from B-RAP lies in the following. If the backoff counter is initially zero or one, the device immediately responds with Msg1 by randomly selecting a sub-channel from among the $2K$ choices in the slot. Otherwise, the device decrements its backoff counter by *two* upon receiving each subsequent slot trigger and responds with Msg1 when the counter reaches zero or one.

To efficiently support operation with $X = 2$, several special cases need clarification:

- If more than K Msg1 responses succeed due to the randomness inherent in sub-channel selection, the reader selects only K devices for Msg3 responses. Excess devices treat this as a Msg1 response failure and wait until the next frame.
- To avoid unnecessary delays, if no responses are detected during the second Msg1 opportunity, the reader terminates it early by immediately transmitting Msg2, as illustrated in Slot 2 of Fig.1(b).
- In the rare event that no devices respond within a slot, the reader briefly waits through two idle Msg1 periods. The second idle period is shortened by early initiation of the next slot, as shown in Slot 3 of Fig.1(b).

D. Proposed Multi-Round Msg1 Collection Random Access Procedure (MRMC-RAP) in A-IoT

To further improve the efficiency of Msg3 responses, we propose an additional operation called the *Multi-Round Msg1 Collection (MRMC)* scheme. The primary objective of MRMC is to maximize utilization of the available K sub-channels by collecting sufficient successful RIDs, thereby ensuring that each slot fully schedules K collision-free Msg3 responses.

Under MRMC, multiple rounds of Msg1 responses can occur within a single slot before the Msg2 transmission. Specifically, after the initial Msg1 response round (triggered by either the paging message or a slot trigger), the reader checks if fewer than K successful RIDs have been collected. If so, it transmits an additional control message called the *backoff trigger*, explicitly designed to initiate subsequent Msg1 response rounds, as illustrated in Fig. 1(c).

Upon receiving a backoff trigger, devices decrement their backoff counters by one. If a device's counter reaches zero, it responds with Msg1 on one of the available K sub-channels. This process repeats until the reader collects at least K successful RIDs or no Msg1 response opportunity can be triggered at the end of the frame. After obtaining K or more successful RIDs, the reader transmits Msg2, scheduling exactly K devices for collision-free Msg3 responses. Excess RIDs beyond K are discarded, and the corresponding devices, which Msg2 does not acknowledge, treat this outcome as a RA failure and wait for the next frame. In the last slot, if the reader collects at least one successful Msg1 response, it still transmits Msg2 for Msg3 scheduling.Therefore, in MRMC-RAP, a device can determine the end of a slot primarily based on the reception of Msg2, as it is followed by a Msg3 response opportunity. As a special case, the reception of a new paging message can also indicate the end of the previous slot and the completion of the previous frame.Considering the dynamic number of Msg1 response opportunities in a slot, the number of slots in a frame is not fixed and can vary during operation.

The proposed MRMC scheme thus effectively maximizes the number of Msg3 responses per slot, potentially improving sub-channel utilization and overall system efficiency.

IV. Performance Analysis

In this section, we analyze the performance of the A-IoT RAP under the assumption that the number of devices awaiting UID reporting, denoted as N, is known in advance at the beginning of each frame. This analysis provides a theoretical foundation for determining the number of Msg1 response opportunities L indicated by the paging message, that maximizes RA efficiency of a frame, defined as the ratio between the expected number of successful Msg3 responses and the frame duration. For practical scenarios, algorithms to estimate N will be introduced later in Section V.

For DL messages, let t_{PG} (bit-duration), t_{ST} (bit-duration) and t_{BT} (bit-duration) represent the time lengths of the paging message, the slot trigger and the backoff trigger transmissions, respectively, and $t_{\mathrm{M2}}(w)$ (bit-duration) denote the transmission duration of Msg2 containing w successfully decoded RIDs. For UL messages, t_{M1} (bit-duration) and t_{M3} (bit-duration) denote the durations of Msg1 and Msg3 responses, respectively. Additionally, if no Msg1 response is detected following a paging message or slot trigger, the reader waits for a short detection period t_{NR} (bit-duration) before issuing a new slot trigger. Here, one bit-duration represents the transmission time of a single bit in the A-IoT system, which is determined during the initialization phase.

For each sub-channel within an Msg1 response opportunity, we define the probabilities as follows: the probability p_0 that no device selects it, the probability p_1 that exactly one device selects it, and the probability p_2 that two or more devices select it. These probabilities are expressed as

$$\begin{aligned} p_0 &= \left(1 - \frac{1}{LK}\right)^N, \\ p_1 &= \frac{N}{LK}\left(1 - \frac{1}{LK}\right)^{N-1}, \\ p_2 &= 1 - p_0 - p_1, \end{aligned} \tag{1}$$

where $\frac{1}{LK}$ denotes the probability that a given device selects a particular sub-channel.

Authorized licensed use limited to: Higher School of Economics. Downloaded on April 07,2026 at 10:42:35 UTC from IEEE Xplore. Restrictions apply.

Sections IV-A and IV-B present the performance analyses of B-RAP and E-RAP, respectively, while Section IV-C analyzes MRMC-RAP. Finally, Section IV-D evaluates the accuracy of the proposed models and derives the optimal L as a function of N.

A. B-RAP

For B-RAP, we first analyze the average duration of a slot and the average number of successful device responses within each slot. Depending on the devices' response behavior, we consider the following three cases:

Case 1: No device selects the slot. The probability of this event, denoted as $P_1^{(\mathrm{B})}$, is given by

$$P_1^{(\mathrm{B})} = p_0^K. \tag{2}$$

As illustrated by the second slot in Fig. 1(a), this slot has duration $T_1^{(\mathrm{B})} = t_{\mathrm{ST}} + t_{\mathrm{NR}}$.

Case 2: Multiple devices select the slot, but all Msg1 responses collide. Thus, all K sub-channels are either idle or collided. The probability of this scenario, denoted as $P_2^{(\mathrm{B})}$, is given by

$$P_2^{(\mathrm{B})} = \sum_{k=1}^{K} \binom{K}{k} p_2^k p_0^{K-k} = (1-p_1)^K - p_0^K. \tag{3}$$

This type of slot has duration $T_2^{(\mathrm{B})} = t_{\mathrm{ST}} + t_{\mathrm{M1}}$.

Case 3: At least one successful Msg1 response occurs, resulting in exactly $w \in \{1, \ldots, K\}$ successful responses. The probability of this event, denoted as $P_3^{(\mathrm{B})}(w)$, is given by

$$P_3^{(\mathrm{B})}(w) = \binom{K}{w} p_1^w (1-p_1)^{K-w}. \tag{4}$$

This type of slot has duration $T_3^{(\mathrm{B})}(w) = t_{\mathrm{ST}} + t_{\mathrm{M1}} + t_{\mathrm{M2}}(w) + t_{\mathrm{M3}}$.

Consequently, the average duration of a slot and the average number of successful Msg3 responses, equal to the number of successful Msg1 responses, can respectively be derived as

$$\begin{aligned} T^{(\mathrm{B})} &= P_1^{(\mathrm{B})} T_1^{(\mathrm{B})} + P_2^{(\mathrm{B})} T_2^{(\mathrm{B})} + \sum_{w=1}^{K} P_3^{(\mathrm{B})}(w) T_3^{(\mathrm{B})}(w) \\ &= t_{\mathrm{ST}} + p_0^K t_{\mathrm{NR}} + (1-p_0^K) t_{\mathrm{M1}} + K p_1 t_{\mathrm{M2}}(1) \\ &\quad + \left[1 - (1-p_1)^K\right] t_{\mathrm{M3}}, \end{aligned} \tag{5}$$

$$S^{(\mathrm{B})} = \sum_{w=1}^{K} w \cdot P_3^{(\mathrm{B})}(w) = K p_1. \tag{6}$$

Finally, the RA efficiency can be derived as

$$R^{(\mathrm{B})} \approx \frac{L \cdot S^{(\mathrm{B})}}{L \cdot T^{(\mathrm{B})}}, \tag{7}$$

where we approximated the duration of t_{PG}, which occurs only in the first slot of the frame, as $t_{\mathrm{PG}} \approx t_{\mathrm{ST}}$.

B. E-RAP

For E-RAP, each slot provides two Msg1 response opportunities. Due to this additional Msg1 opportunity compared to B-RAP, more device-response scenarios can occur within each slot, summarized as follows:

Case 1: No successful Msg1 response occurs in the first Msg1 response opportunity (no one transmits or all Msg1 responses collide), and no device send Msg1 in the second Msg1 response opportunity. The probability of this event, denoted as $P_1^{(\mathrm{E})}$, is

$$P_1^{(\mathrm{E})} = (1-p_1)^K \cdot p_0^K. \tag{8}$$

As illustrated by the third slot in Fig. 1(b), this slot has duration $T_1^{(\mathrm{E})} = t_{\mathrm{ST}} + t_{\mathrm{M1}} + t_{\mathrm{NR}}$.

Case 2: There exist Msg1 responses in the second Msg1 response opportunity, but all Msg1 responses in this slot collide. The probability of this event, denoted as $P_2^{(\mathrm{E})}$, is

$$P_2^{(\mathrm{E})} = (1-p_1)^K [(1-p_1)^K - p_0^K]. \tag{9}$$

This type of slot has duration $T_2^{(\mathrm{E})} = t_{\mathrm{ST}} + 2t_{\mathrm{M1}}$.

Case 3: At least one successful Msg1 response occurs in the first Msg1 response opportunity, resulting in exactly $w \in \{1, \ldots, K\}$ successful responses, but no device send Msg1 in the second Msg1 response opportunity. The probability of this event, denoted as $P_3^{(\mathrm{E})}(w)$, is

$$P_3^{(\mathrm{E})}(w) = \left[\binom{K}{w} p_1^w (1-p_1)^{K-w}\right] p_0^K. \tag{10}$$

As illustrated by the second slot in Fig. 1(b), this slot has duration $T_3^{(\mathrm{E})}(w) = t_{\mathrm{ST}} + t_{\mathrm{M1}} + t_{\mathrm{NR}} + t_{\mathrm{M2}}(w) + t_{\mathrm{M3}}$.

Case 4: There exist Msg1 responses in the second Msg1 response opportunity, and at least one successful Msg1 response occurs in this slot. The exact number of successful responses in the two Msg1 response opportunities are $w_1 \in \{0, \ldots, K\}$ and $w_2 \in \{0, \ldots, K\}$, respectively, while $w_1 + w_2 \geq 1$.

When $w_2 = 0$, $w_1 > 0$. It means that the Msg1 response occurs in the second Msg1 response opportunity, but all collide. Thus, the probability of this event, denoted as $P_{41}^{(\mathrm{E})}(w_1)$, is

$$P_{41}^{(\mathrm{E})}(w_1) = \left[\binom{K}{w_1} p_1^{w_1} (1-p_1)^{K-w_1}\right] \cdot [(1-p_1)^K - p_0^K]. \tag{11}$$

This type of slot has duration $T_{41}^{(\mathrm{E})}(w_1) = t_{\mathrm{ST}} + 2\ t_{\mathrm{M1}} + t_{\mathrm{M2}}(w_1) + t_{\mathrm{M3}}$.

When $w_2 > 0$, $w_1 \geq 0$. The probability of this event, denoted as $P_{42}^{(\mathrm{E})}(w_1, w_2)$, is

$$\begin{aligned} P_{42}^{(\mathrm{E})}(w_1, w_2) &= \binom{K}{w_1} p_1^{w_1} (1-p_1)^{K-w_1} \\ &\quad \cdot \binom{K}{w_2} p_1^{w_2} (1-p_1)^{K-w_2}. \end{aligned} \tag{12}$$

However, considering that the Msg2 carries at most K RIDs, this type of slot has duration $T_{42}^{(\mathrm{E})}(w_1, w_2) = t_{\mathrm{ST}} + 2\ t_{\mathrm{M1}} + t_{\mathrm{M2}}\left(\min\{w_1 + w_2, K\}\right) + t_{\mathrm{M3}}$. Meanwhile, there are at most K successful Msg3 responses even for $w_1 + w_2 > K$.

Consequently, the average duration of a slot and the average number of successful Msg3 responses can respectively be derived as

$$\begin{aligned} T^{(\mathrm{E})} &= P_1^{(\mathrm{E})} \cdot T_1^{(\mathrm{E})} + P_2^{(\mathrm{E})} \cdot T_2^{(\mathrm{E})} \\ &\quad + \sum_{w=1}^{K} P_3^{(\mathrm{E})}(w) \cdot T_3^{(\mathrm{E})}(w) \end{aligned}$$

Authorized licensed use limited to: Higher School of Economics. Downloaded on April 07,2026 at 10:42:35 UTC from IEEE Xplore. Restrictions apply.

$$+\sum_{w_1=1}^{K} P_{41}^{(\mathrm{E})}(w_1)\cdot T_{41}^{(\mathrm{E})}(w_1)$$
$$+\sum_{w_2=1}^{K}\sum_{w_1=0}^{K} P_{42}^{(\mathrm{E})}(w_1,w_2)\cdot T_{42}^{(\mathrm{E})}(w_1,w_2),$$
$$S^{(\mathrm{E})}=\sum_{w=1}^{2K}\min\{w,K\}\cdot\binom{2K}{w}p_1^w(1-p_1)^{2K-w}. \quad (13)$$

Finally, the RA efficiency can be derived as

$$R^{(\mathrm{E})}\approx\frac{L\cdot S^{(\mathrm{E})}}{L\cdot T^{(\mathrm{E})}}, \quad (14)$$

where we approximated the duration of t_{PG} as t_{ST} for simplicity.

C. MRMC-RAP

A unique property of MRMC-RAP is that the number of Msg1 response opportunities may vary from slot to slot. This subsection presents an analysis considering this characteristic.

Firstly, the number of successful Msg1 in one Msg1 response opportunity, denoted by a random variable W, follows a binomial distribution $\mathrm{Bin}(K,p_1)$, which means

$$P\{W=w\}=\binom{K}{w}p_1^w(1-p_1)^{K-w},\ w=0,\ldots,K. \quad (15)$$

After l (≥ 1) Msg1 responses opportunities, the number of successful Msg1 becomes the sum of a series of random variables denoted by $W^{(l)}=\sum_{i=1}^{l}W_i$ where W_i represents the number of successful Msg1 in i-th Msg1 responses opportunity. Due to the independence of device behavior between Msg1 response opportunities, $W^{(l)}$ follows the binomial distribution $\mathrm{Bin}(Kl,p_1)$, where

$$P\{W^{(l)}=w\}=\binom{Kl}{w}p_1^w(1-p_1)^{Kl-w},\ w=0,\ldots,Kl. \quad (16)$$

Then, note the random variable Y as the number of Msg1 response opportunities that the reader first collects at least K successful Msg1. For $Y=1$, the reader collects K successful Msg1 responses in the first Msg1 response opportunity, and the probability for this event is

$$P\{Y=1\}=P(W\geq K)=p_1^K. \quad (17)$$

For $Y=l>1$, it means that the reader collects at most $K-1$ successful RIDs during the first $(l-1)$ Msg1 response opportunities, and in the l-th Msg1 response opportunity, there are at least K successful RIDs. Thus, the probability of this event is

$$\begin{aligned}P\{Y=l\}&=P\{W^{(l-1)}\leq K-1, W^{(l)}\geq K\}\\&=\sum_{w_1=0}^{K-1}P\{W^{(l-1)}=w_1\}\cdot\sum_{w_2=K-w_1}^{K}P\{W=w_2\}\\&=\sum_{w_1=0}^{K-1}\sum_{w_2=K-w_1}^{K}\binom{Kl-K}{w_1}\binom{K}{w_2}p_1^{w_1+w_2}\\&\quad\cdot(1-p_1)^{Kl-(w_1+w_2)}. \quad (18)\end{aligned}$$

Since each Y corresponds to the number of required Msg1 response opportunities in one slot, we further denote

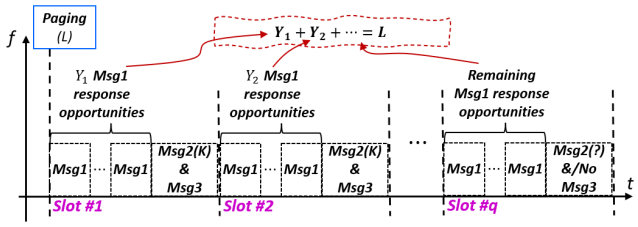


Fig. 2. Stochastic renewal process within a frame initiated by paging with L.

$Y^{(m)}=\sum_{j=1}^{m}Y_j$ as the required Msg1 response opportunities of m slots, where Y_j represents the number of required Msg1 response opportunities in j-th slot. The corresponding probability is

$$\begin{aligned}P\{Y^{(m)}=l\}&=\sum_{j=m-1}^{l-1}P\{Y^{(m-1)}=j, Y=l-j\}\\&=\sum_{j=m-1}^{l-1}P\{Y^{(m-1)}=j\}\cdot P\{Y=l-j\}.\end{aligned} \quad (19)$$

As illustrated in Fig. 2, the transmissions within the frame could be seen as a stochastic renewal process. Denote the random variable Q as the number of slots in a frame initiated by the paging message with parameter L, where $1\leq Q\leq L$. For $Q=1$, it means that after L Msg1 response opportunities, the reader collects at most K successful Msg1 responses. So the corresponding probability is

$$P\{Q=1\}=P\{Y\geq L\}=1-\sum_{j=1}^{L-1}P\{Y=j\}. \quad (20)$$

For $Q=q>1$, as shown in Fig. 2, the reader can collect at least K successful RIDs in each of the first $(q-1)$ slots, but this cannot be guaranteed in the last slot. To address this case, there must be $Y^{(q-1)}<L$ and $Y^{(q)}\geq L$, which means less than L Msg1 response opportunities are used among the first $(q-1)$ slots and the rest of the Msg1 response opportunities are left for the q-th slot. These Msg1 response opportunities may be enough for collecting K successful Msg1 responses, or may not. So the probability could be

$$P\{Q=q\}=\sum_{j_1=q-1}^{L-1}\sum_{j_2=L-j_1}^{L}P\{Y^{(q-1)}=j_1\}\cdot P\{Y=j_2\}. \quad (21)$$

Finally, we can discuss the RA efficiency of the entire frame under different values of Q. When $Q=1$, the number of expected successful Msg3 responses in this frame is

$$\begin{aligned}S_1^{(\mathrm{M})}&=K\cdot P\{Y=L|Q=1\}\\&\quad+\sum_{w=0}^{K-1}w\cdot P\{W^{(L)}=w|Q=1\}\\&=K\cdot\frac{P\{Y=L\}}{P\{Q=1\}}+\sum_{w=0}^{K-1}w\cdot\frac{P\{W^{(L)}=w\}}{P\{Q=1\}}.\end{aligned} \quad (22)$$

Authorized licensed use limited to: Higher School of Economics. Downloaded on April 07,2026 at 10:42:35 UTC from IEEE Xplore. Restrictions apply.

The corresponding frame length is

$$T_1^{(\mathrm{M})} = t_{\mathrm{M1Round}} \cdot L + [t_{\mathrm{M2}}(K) + t_{\mathrm{M3}}] \frac{P\{Y = L\}}{P\{Q = 1\}} + \sum_{w=1}^{K-1} [t_{\mathrm{M2}}(w) + t_{\mathrm{M3}}] \frac{P\{W^{(L)} = w\}}{P\{Q = 1\}}, \quad (23)$$

where t_{M1Round} represents the expected length of one slot trigger or backoff trigger plus one Msg1 response opportunity. Here, the duration of t_{BT} is approximated as $t_{\mathrm{BT}} \approx t_{\mathrm{ST}}$, since the design of the backoff trigger is similar to the slot trigger. So we have $t_{\mathrm{M1Round}} \approx t_{\mathrm{ST}} + \left[p_0^K \cdot t_{\mathrm{NR}} + (1 - p_0^K) t_{\mathrm{M1}}\right]$.

When $Q = q \geq 2$, the expected successful Msg3 responses in this frame can be obtained as

$$S_q^{(\mathrm{M})} = qK \cdot \frac{P\{Y^{(q)} = L\}}{P\{Q = q\}} + \sum_{l=q-1}^{L-1} \sum_{w=0}^{K-1} [(q-1)K + w] \cdot \frac{P\{Y^{(q)} = l\} \cdot P(W^{(L-l)} = w)}{P\{Q = q\}}. \quad (24)$$

The first term in (24) represents that at least K successful Msg1 responses are collected in the last slot, whereas the second term represents the case that there are at most $(K-1)$ successful Msg1 responses in the last slot.

Meanwhile, the corresponding frame length can also be calculated as

$$\begin{aligned} T_q^{(\mathrm{M})} = {} & t_{\mathrm{M1Round}} \cdot L + [q \cdot K \cdot t_{\mathrm{M2}} + q \cdot t_{\mathrm{M3}}] \frac{P\{Y^{(q)} = L\}}{P\{Q = q\}} \\ & + \sum_{l=q-1}^{L-1} [(q-1)K \cdot t_{\mathrm{M2}} + (q-1) t_{\mathrm{M3}}] \cdot \frac{P\{Y^{(q-1)} = l\} \cdot P(W^{(L-l)} = 0)}{P\{Q = q\}} \\ & + \sum_{l=q-1}^{L-1} \sum_{w=1}^{K-1} [((q-1)K + w)\, t_{\mathrm{M2}} + q \cdot t_{\mathrm{M3}}] \cdot \frac{P\{Y^{(q-1)} = l\} \cdot P(W^{(L-l)} = w)}{P\{Q = q\}}. \end{aligned} \quad (25)$$

The second term in (25) corresponds to the event in which the reader collects at least K RIDs during the q-th slot. The third term captures the case where no Msg1 responses are successfully decoded in that slot without Msg3 response scheduling, whereas the fourth term accounts for the intermediate situation in which the number of successful Msg1 responses lies between 1 and $K-1$ and the reader still needs to schedule a Msg3 response for this last slot.

Consequently, the average duration of the frame and the average number of successful Msg3 responses can respectively be derived as

$$\begin{aligned} T^{(\mathrm{M})} &= \sum_{q=1}^{L} T_q^{(\mathrm{M})} \cdot P\{Q = q\}, \\ S^{(\mathrm{M})} &= \sum_{q=1}^{L} S_q^{(\mathrm{M})} \cdot P\{Q = q\}. \end{aligned} \quad (26)$$

Finally, the RA efficiency can be derived as

$$R^{(\mathrm{M})} \approx \frac{S^{(\mathrm{M})}}{T^{(\mathrm{M})}}, \quad (27)$$

TABLE II
THE PROPORTIONALITY CONSTANT α

Scheme \ K	1	2	3	4	5	6
B-RAP	1.42	1.13	1.04	1.01	1.00	1.00
E-RAP	1.22	1.07	1.02	1.02	1.00	1.00
MRMC-RAP	1.42	1.34	1.18	1.09	1.07	1.04

where we still approximated the duration of t_{PG} as t_{ST} for simplicity.

D. Optimal L Values

We first evaluate the accuracy of the proposed analytical models for the three RAP schemes. For the simulation scenario, we set the paging message duration $t_{\mathrm{PG}} = 22$ bit-durations; the durations of the slot trigger and the backoff trigger $t_{\mathrm{ST}} = t_{\mathrm{BT}} = 4$ bit-durations; Msg1 duration $t_{\mathrm{M1}} = 16$ bit-durations; Msg3 duration $t_{\mathrm{M3}} = 96$ bit-durations, significantly longer than t_{M1}; Msg2 duration $t_{\mathrm{M2}}(w) = 16w$ bit-durations, where w denotes the number of successful Msg1 responses within a slot; and the no-response interval $t_{\mathrm{NR}} = 4$ bit-durations. As 3GPP standardization is still ongoing and these parameters are not yet finalized, we have reasonably selected these values by referencing typical RFID system parameters. However, our analytical models remain applicable and effective for different parameter sets as well.

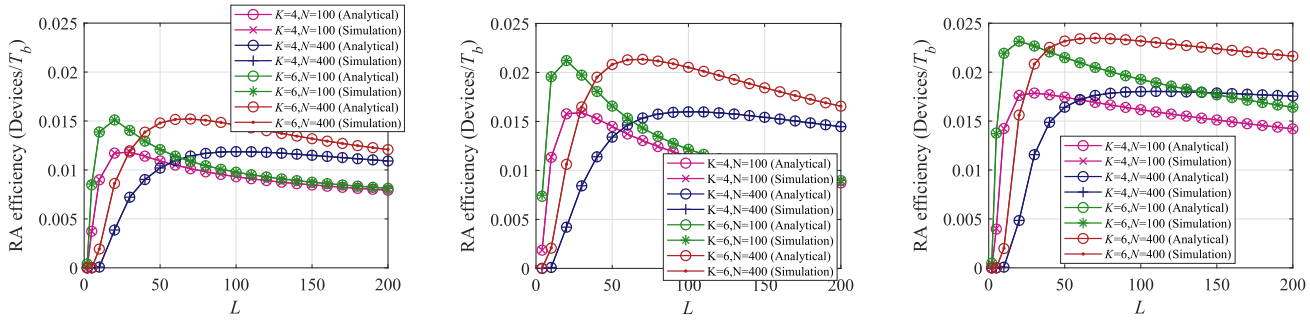
Fig. 3 illustrates the RA efficiency results for varying L values with $K = 4$ or 6, and $N = 100$ or 400. For each simulation scenario, we run 10^4 frames to ensure statistically reliable results. As shown, the analytical results closely align with the simulation results, confirming the effectiveness of our analytical models. MRMC-RAP generally achieves the highest RA efficiency, whereas B-RAP performs the worst. For instance, when $K = 6$ and $N = 400$, MRMC-RAP achieves a 54% higher RA efficiency than B-RAP, while E-RAP achieves a 40% improvement over B-RAP.

Based on the observations from Fig. 3, there exists an optimal number of slots per frame, denoted by L^*, that maximizes the RA efficiency for each scheme. To further investigate this relationship, Fig. 4(a) shows the optimal L^* values with varying N. As shown, L^* increases proportionally with N. Additionally, though not explicitly shown here, we observed that L^* decreases inversely proportional to K. Motivated by these findings, we propose a linear relationship defined by

$$L^* = \alpha \cdot \frac{N}{K}, \quad (28)$$

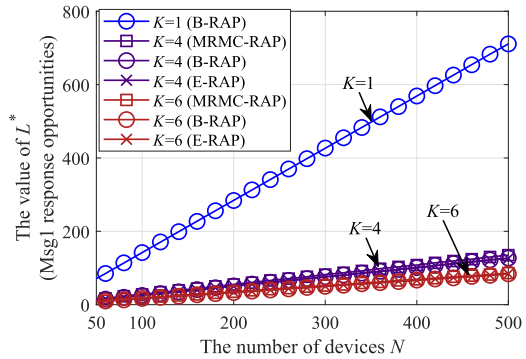
where α denotes a proportionality constant. Table II summarizes the obtained α values for the three RAP schemes under various K settings. Interestingly, while α significantly deviates from 1 when $K = 1$, it converges toward 1 as K increases. Consequently, if the number of devices N is known during A-IoT network operation, the optimal L^* value can be directly computed using (28). In addition, Fig. 4(b) shows the maximum RA efficiency as a function of the number of devices N. Interestingly, the maximum RA efficiency remains nearly constant and independent of N. For example, when $K = 4$, the

Authorized licensed use limited to: Higher School of Economics. Downloaded on April 07,2026 at 10:42:35 UTC from IEEE Xplore. Restrictions apply.

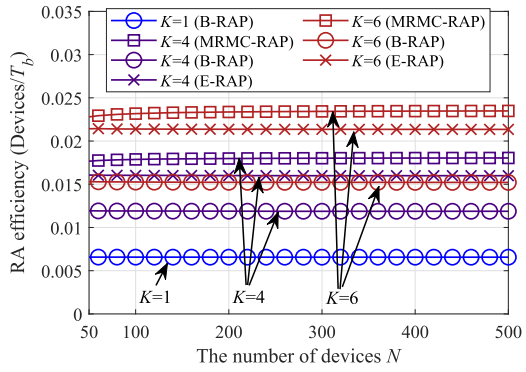


(a) B-RAP (b) E-RAP (c) MRMC-RAP

Fig. 3. RA efficiency versus L.



(a) The optimal L^* versus N.



(b) Maximum RA efficiency versus N

Fig. 4. Optimal L^* and corresponding maximum RA efficiency versus the number of devices (N).

corresponding maximum RA efficiencies are approximately 0.012, 0.016, and 0.018 for B-RAP, E-RAP, and MRMC-RAP, respectively. It is noted that these maximum values are obtained when the optimal L^* is configured based on N. As E-RAP and MRMC-RAP are designed for multi-channel operations, only B-RAP is applied in the single-channel case ($K = 1$).

1) Capture Effect in Backscatter-Based A-IoT Systems: The capture effect is a well-known phenomenon in wireless communication, where a transmission can still be decoded when multiple devices transmit simultaneously, provided that its received power is significantly higher than that of the others. In backscatter-based A-IoT systems, this effect becomes more pronounced because the reader emits a CW that is reflected by devices and received again at the reader. The resulting round-trip propagation introduces a double path loss, which further amplifies power disparities among devices and thereby enhances the likelihood of capture [36].

A similar phenomenon has also been extensively investigated in RFID systems, which likewise operate using backscatter communication. In this context, the capture effect has been analyzed in recent studies [18], [36] and is often characterized by a capture probability, denoted as p_{ca}, representing the likelihood of successfully decoding one signal within a collision when multiple devices transmit their RIDs simultaneously.

Considering the capture effect, the success probability of decoding a Msg1 response at the reader can be expressed as

$$p'_S = \frac{N}{LK}\left(1-\frac{1}{LK}\right)^{N-1} + p_{\text{ca}}\left[1-\left(1-\frac{1}{LK}\right)^{N} - \frac{N}{LK}\left(1-\frac{1}{LK}\right)^{N-1}\right]. \quad (29)$$

By repeating a similar analysis to that in Section IV-D while incorporating the effect of p_{ca}, we still obtain a linear relationship that approximates the optimal L^*_{ca} as a function of the number of subchannels K and devices N, given by

$$L^*_{\text{ca}} = \alpha \cdot (1-p_{\text{ca}}) \cdot \frac{N}{K}, \quad (30)$$

where α still takes the value from Table II.

In practice, the value of p_{ca} depends on the deployment scenario and the spatial distribution of devices, and thus cannot be precisely determined by the reader in advance. In inventory scenarios, in particular, the spatial distribution of devices may vary significantly. Consequently, the algorithms proposed in this paper do not rely on any prior knowledge of p_{ca}. However, if a genie-aided ideal algorithm were to know p_{ca} and utilize the optimal L^*_{ca} given in (30), it would provide an upper-bound on performance. Furthermore, if our proposed algorithms, which do not use p_{ca}, achieve comparable performance to this ideal case, their effectiveness can be validated.

V. Proposed Online Control Algorithms

In A-IoT networks, the number of Msg1 response opportunities L broadcast by the reader significantly impacts RA efficiency. An excessively small L causes frequent Msg1 collisions, while an excessively large L leads to under-utilization

Authorized licensed use limited to: Higher School of Economics. Downloaded on April 07,2026 at 10:42:35 UTC from IEEE Xplore. Restrictions apply.

of available sub-channels. Although the optimal L can be calculated using (28), this approach requires prior knowledge of the number of devices awaiting UID reporting at the beginning of each frame, denoted by N, which is typically unavailable in practice. It is important to note that an efficient algorithm must not only accurately estimate the number of devices in the network but also do so rapidly to minimize inefficient RAP operations. This estimation becomes particularly challenging when N is large, that is, under highly bursty traffic conditions. To address this, we propose two online control algorithms that dynamically estimate N based on observed channel outcomes and adaptively adjust L for each subsequent frame:

- *Basic Estimation (BEst)*, introduced in Section V-A, estimates the number of devices by observing the number of collided sub-channels from the previous frame and multiplying it by the average number of devices per collision. This approach extends conventional RFID estimation methods [19] to accommodate the multi-channel nature of A-IoT networks.
- *Boosted Estimation with Early Frame Termination (BE^{2}FT)*, introduced in Section V-B, further enhances accuracy by explicitly considering bursty IoT traffic conditions, where the average number of devices involved in collisions can increase significantly. Additionally, it incorporates an early frame termination mechanism to limit RA inefficiencies within the current frame caused by estimation inaccuracies.

A. Basic Estimation (BEst)

Suppose there are N devices contending at the start of a frame, but the reader's estimation is $\hat{N}$. Without exact knowledge of N, the reader determines L based on $\alpha \cdot \frac{\hat{N}}{K}$ according to (28). At the end of the frame, the reader observes the total number of sub-channels experiencing Msg1 collisions, denoted as C. If n_c represents the average number of devices involved per collided sub-channel, the number of devices contending in the next frame, $\hat{N}_{\text{next}}$, can be estimated as:

$$\hat{N}_{\text{next}} = C \cdot n_c. \tag{31}$$

Next, we estimate the value of n_c. For this purpose, we first derive the probability $P_c(m)$ that exactly m devices select the same sub-channel within a single Msg1 response opportunity:

$$\begin{aligned} P_c(m) &= \binom{N}{m}\left(\frac{1}{LK}\right)^m\left(1-\frac{1}{LK}\right)^{N-m} \\ &= \binom{N}{m}\left(\frac{1}{\alpha\hat{N}}\right)^m\left(1-\frac{1}{\alpha\hat{N}}\right)^{N-m}, \end{aligned} \tag{32}$$

where we applied the relationship $L = \alpha \cdot \frac{\hat{N}}{K}$. Consequently, we have

$$\begin{aligned} n_c &= \sum_{m=2}^{N} m \cdot P\{m|\text{Collision}\} \\ &= \sum_{m=2}^{N} m \cdot \frac{P_c(m)}{\sum_{j=2}^{N} P_c(j)} \\ &= \frac{\frac{N}{\alpha\hat{N}}\left[1-\left(1-\frac{1}{\alpha\hat{N}}\right)^{N-1}\right]}{1-\left(1-\frac{1}{\alpha\hat{N}}\right)^{N}-\frac{N}{\alpha\hat{N}}\left(1-\frac{1}{\alpha\hat{N}}\right)^{N-1}}. \end{aligned} \tag{33}$$

If the estimate $\hat{N}$ is close to the true device number N, assuming large $\hat{N}$, we apply the limit $\lim_{n\to\infty}(1-1/(\alpha n))^n = e^{-1/\alpha}$ and approximate n_c as:

$$n_c \approx \frac{1-e^{-1/\alpha}}{\alpha-(1+\alpha)e^{-1/\alpha}}. \tag{34}$$

For example, when $\alpha = 1$, we obtain $n_c \approx 2.3922$.

In summary, upon observing C collided sub-channels within a frame, the reader estimates the device number for the next frame simply as $\hat{N}_{\text{next}} = C \cdot n_c$, where n_c is computed by (34), and accordingly determines the next frame length L. It should be noted that Basic Estimation is accurate only when the discrepancy between $\hat{N}$ and the actual N is not substantial.

B. Boosted Estimation With Early Frame Termination (BE^{2}FT)

In bursty or sparse IoT traffic scenarios, the initial estimate $\hat{N}$ may significantly differ from the actual number of devices N, reducing the accuracy of BEst. To address this limitation, we propose BE^{2}FT, consisting of two main components: 1) a correction step for $\hat{N}$, considering that the actual number of devices per collided sub-channel can be substantially higher or lower than n_c computed by (34); and 2) an early frame termination mechanism to mitigate resource waste caused by inaccurate estimations of $\hat{N}$.

1) Boosted Estimation Through Correcting $\hat{N}$: Remind that the reader only has the estimated $\hat{N}$ based on which L is determined by $\alpha \cdot \frac{\hat{N}}{K}$ and it does not have the value of the exact N. To quantify the difference between $\hat{N}$ and N, we introduce a difference factor β defined by

$$N = \beta\hat{N}. \tag{35}$$

A value of β closer to 1 indicates higher estimation accuracy. To correct $\hat{N}$, we determine an appropriate β by observing the channel outcomes within each frame.

At the end of each frame, the reader observes the total number of successful Msg3 responses, denoted as S, and the total number of sub-channels experiencing Msg1 collisions, denoted as C. The expected value of C is then calculated as:

$$\begin{aligned} E[C] &= LK\left[1-\left(1-\frac{1}{LK}\right)^N-\binom{N}{1}\frac{1}{LK}\left(1-\frac{1}{LK}\right)^{N-1}\right] \\ &= \alpha\hat{N}-\alpha\hat{N}\left(1-\frac{1}{\alpha\hat{N}}\right)^{\beta\hat{N}}-\beta\hat{N}\left(1-\frac{1}{\alpha\hat{N}}\right)^{\beta\hat{N}-1}, \end{aligned} \tag{36}$$

where we applied the relations $LK = \alpha\hat{N}$ and $N = \beta\hat{N}$.

On the other hand, if the estimate $\hat{N}$ is accurate (i.e., $\beta \approx 1$), the expected value of C observed by the reader can be expressed as:

$$E[\hat{C}] = \alpha\hat{N}-\alpha\hat{N}\left(1-\frac{1}{\alpha\hat{N}}\right)^{\hat{N}}-\hat{N}\left(1-\frac{1}{\alpha\hat{N}}\right)^{\hat{N}-1}. \tag{37}$$

Authorized licensed use limited to: Higher School of Economics. Downloaded on April 07,2026 at 10:42:35 UTC from IEEE Xplore. Restrictions apply.

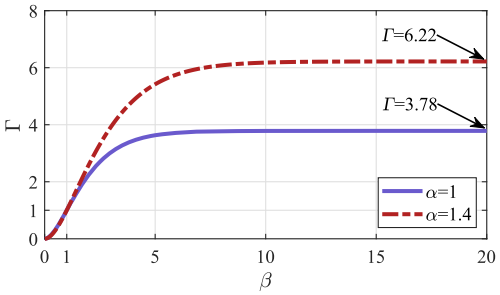


Fig. 5. Ratio of actual to estimated collisions, $\Gamma = \frac{E[C]}{E[\hat{C}]}$, versus ratio of actual to estimated devices, $\beta = \frac{N}{\hat{N}}$.

Consequently, the difference between (36) and (37) implicitly indicates the accuracy level of the estimation. Let Γ denote their ratio, which can be approximated as:

$$\Gamma = \frac{E[C]}{E[\hat{C}]} \approx \frac{\alpha - (\alpha+\beta)e^{-\beta/\alpha}}{\alpha - (\alpha+1)e^{-1/\alpha}}, \tag{38}$$

where the approximation assumes that $\hat{N}$ tends to infinity. Fig. 5 shows the Γ values for different combinations of α and β.

In practice, since the reader cannot directly calculate $E[C]$ due to the unknown N, it approximates $E[C]$ by the observed number of collided sub-channels, C. Thus, the reader estimates Γ as $\frac{C}{E[\hat{C}]}$ and then inversely derives β from Γ using (38). The estimation can then be refined by updating $\hat{N}$ to $\beta\hat{N}$. Furthermore, by accounting for the number of successful Msg3 responses S, the estimated number of devices for the subsequent frame is given by:

$$\hat{N}_{\text{next}} = \beta\hat{N} - S. \tag{39}$$

2) Early Frame Termination: If a frame begins with an estimate $\hat{N}$ significantly deviating from the actual device number N, running the entire frame may result in inefficient channel utilization. To address this inefficiency, it is practical to implement early frame termination by sending a new paging message. Similar approaches have been widely studied in conventional RFID systems [17], [21].

Considering the unique characteristics of A-IoT, we propose the following criteria for triggering early frame termination. Specifically, we evaluate conditions at approximately one-quarter of the total Msg1 response opportunities, i.e., $\frac{L}{4}$. Note that the value $\frac{1}{4}$ was empirically determined through trial-and-error experiments. Let C' denote the number of sub-channels experiencing Msg1 collisions up to this point. The conditions for early frame termination are:

- **Excessive collisions:** If $C' > 1.2 \times \frac{E[C]}{4}$, i.e., collisions are significantly higher (by more than 20%) than the expected number if the estimation is accurate. This scenario indicates that the actual number of devices N considerably exceeds the estimated $\hat{N}$.
- **Too few collisions:** If $C' < 0.8 \times \frac{E[C]}{4}$, i.e., collisions are notably fewer (by more than 20%) than the expected number if the estimation is accurate. This scenario indicates that the actual number of devices N is considerably lower than the estimated $\hat{N}$, potentially causing severe under-utilization of radio resources if the current frame continues.

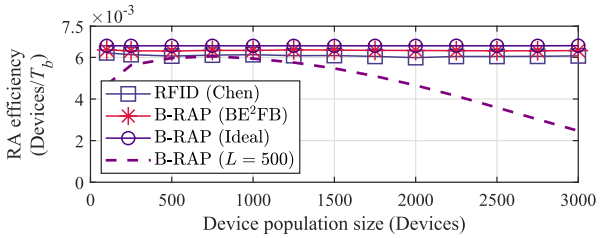
Algorithm 1 Boosted Estimation With Early Frame Termination

1: Initialize K, α according to Table II, $\hat{N}_1 = 100$, $L_1 = \lceil \alpha\hat{N}_1/K \rceil$. Calculate n_C according to (34).
2: In the j-th frame with L_j, the reader calculate $E[\hat{C}_j]$ according to (37).
3: **if** the reader has allocated $\left\lceil \frac{L_j}{4} \right\rceil$ Msg1 response opportunities **then**
4: The reader counts the number of Msg1 collisions as C'_j and the number of Msg3 responses as S'_j.
5: **if** $C'_j > 1.2 \times \frac{E[\hat{C}_j]}{4}$ or $C'_j < 0.8 \times \frac{E[\hat{C}_j]}{4}$ **then**
6: The reader calculate β_j based on (38).
7: $\hat{N}_{j+1} = \beta_j \hat{N}_j - S'_j$.
8: $L_{j+1} = \left\lceil \alpha \frac{\hat{N}_{j+1}}{K} \right\rceil$.
9: The reader broadcast a new paging with L_{j+1}. Go to Line 2.
10: **end if**
11: **end if**
12: The reader continues the remaining part of frame j.
13: **if** no Msg1 response or no collision in j-th frame **then**
14: The reader ends the RAP.
15: **else**
16: The reader counts the number of Msg1 collisions as C_j and the number of Msg3 responses as S_j.
17: The reader calculate β_j based on (38).
18: $\hat{N}_{j+1} = \beta_j \hat{N}_j - S_j$.
19: $L_{j+1} = \left\lceil \alpha \frac{\hat{N}_{j+1}}{K} \right\rceil$.
20: The reader broadcast a new paging with L_{j+1}. Go to Line 2.
21: **end if**

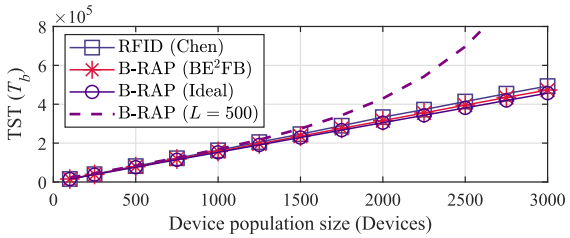
3) Algorithm Description: By integrating the proposed correction of estimation, the early frame termination mechanism, and the optimal L determination from Section IV-D, we can finally present the overall architecture of our online control algorithm, BE^{2}FT, in Algorithm 1.It is worth noting that the reader is the entity that performs Algorithm 1, which estimates the number of devices, determines the parameter L for the new frame, and decides whether to trigger early frame termination. Each device relies solely on the reader's guidance, such as the paging message, to perform simple UL responses.

As shown in Algorithm 1, Line 1 initializes key parameters. Lines 2 and 4 prepare for the early frame termination decision. Once the RAP reaches one-quarter of the frame in Line 3, the triggering of early frame termination relies on the 'excessive collisions' and 'too few collisions' conditions in Line 5. If early frame termination is triggered, Lines 6-7 describe the procedures for boosted estimation; the reader calculates the L_{j+1} in Line 8 and broadcasts it to initiate a new frame in Line 9. Note that S'_j represents the number of successful Msg3 responses at one-quarter of the total Msg1 response opportunities, and it should be accounted for in the refined estimation $\beta_j\hat{N}_j$ in Line 7, as well as in Line 18. If early frame

Authorized licensed use limited to: Higher School of Economics. Downloaded on April 07,2026 at 10:42:35 UTC from IEEE Xplore. Restrictions apply.

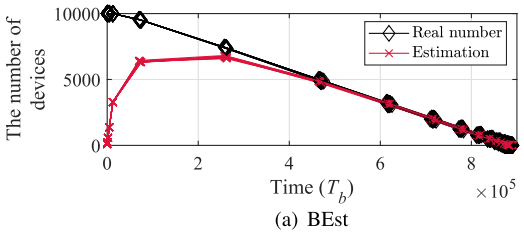


(a) RA efficiency versus device population size.

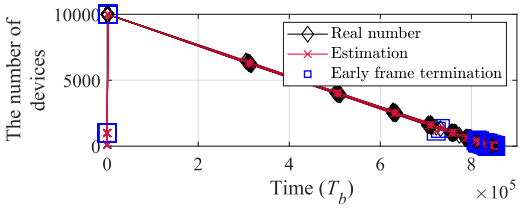


(b) Total service time (TST) versus device population size.

Fig. 6. RA efficiency and total service time (TST) for different RA control algorithms in a single-channel system ($K = 1$).



(a) BEst



(b) BE^{2}FT

Fig. 7. Estimation performance with $K = 4$ sub-channels and 10000 devices.

termination is not triggered, the frame continues. Once the frame ends, if the reader does not observe any Msg1 response or collision as in Line 13, the reader ends the RAP in Line 14. If collisions still exist, the reader continues to perform boosted estimation in Lines 16-18, calculates L_{j+1} accordingly in Line 19, and broadcasts it to indicate the next frame in Line 20.

VI. Performance Evaluation

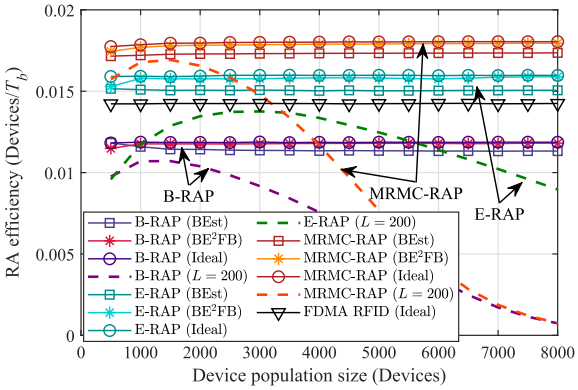
In this section, we present simulation results for the three RAP schemes, B-RAP, E-RAP, and MRMC-RAP, combined with the proposed online control algorithms, BEst and BE^{2}FT. For a better discussion, we first define two performance metrics used in the following evaluations: the total service time (TST) and the RA efficiency:

- *Total service time (TST):* It is defined as the duration from the time the reader sends the first paging for starting the inventory process to the time when all devices successfully finish UID reporting. Consequently, multiple frames may occur within this duration.
- *RA efficiency:* It is defined as the ratio between the number of overall successful Msg3 responses and the TST among the entire inventory process.

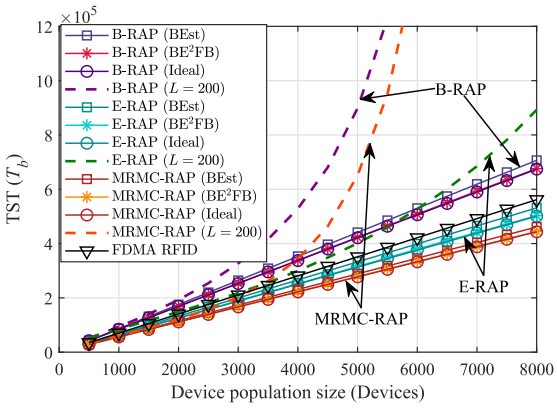
These two metrics are inversely related; lower RA efficiency leads to a higher TST value, and vice versa.For the evaluations, we simulate different values of the *device population size*, which denotes the number of devices present at the beginning of the inventory process to report their UIDs. Consequently, over one complete inventory process, multiple frames are required for all devices in the population to successfully report their UIDs.Table III summarizes the key simulation configurations used in this section. Unless otherwise specified, the same settings are used throughout the evaluations.

A. Performance Comparison in Single-Channel A-IoT System

We first consider the single-channel A-IoT system ($K = 1$) with B-RAP, similar to the conventional RFID system, where each paging or slot trigger can initiate only one Msg1 response opportunity of each slot.



(a) RA efficiency versus device population size.



(b) Total service time (TST) versus device population size.

Fig. 8. RA efficiency and total service time (TST) for different RA control algorithms in a multi-channel system with $K = 4$ sub-channels.

For better comparison, in addition to BE^{2}FT, we introduce the following two algorithms:

- **Ideal algorithm:** This algorithm assumes that the reader perfectly knows the actual number of devices at any time and performs RA control accordingly. This algorithm serves as the performance upper bound.

Authorized licensed use limited to: Higher School of Economics. Downloaded on April 07,2026 at 10:42:35 UTC from IEEE Xplore. Restrictions apply.

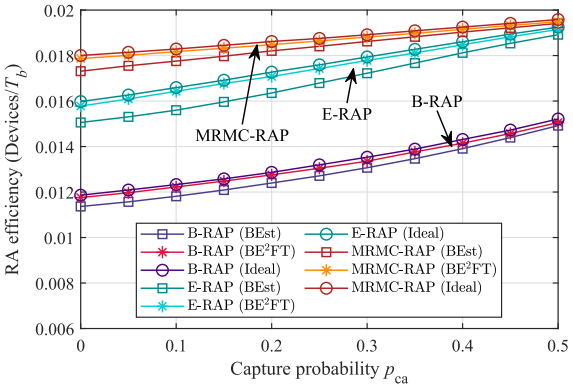


Fig. 9. RA efficiency versus capture probability p_{ca} with $K = 4$ sub-channels and device population size 3000.

TABLE III
SIMULATION CONFIGURATIONS

Parameter	Value
[Network configurations]	
Sub-channel count K	1 (single-channel system); 4 (multi-channel system)
Paging message duration t_{PG}	22 bit-durations
Slot trigger duration t_{ST}	4 bit-durations
Backoff trigger duration t_{BT}	4 bit-durations
Msg1 duration t_{M1}	16 bit-durations
Msg2 duration $t_{\text{M2}}(w)$	$16w$ bit-durations
Msg3 duration t_{M3}	96 bit-durations
No-response interval t_{NR}	4 bit-durations
Capture probability p_{ca}	0 by default; varied as specified (Fig. 9)
[Algorithm configurations]	
Proportionality constant α	1.42 ($K = 1$, B-RAP); 1.01 ($K = 4$, B-RAP); 1.02 ($K = 4$, E-RAP); 1.09 ($K = 4$, MRMC-RAP)
[Evaluation setup]	
Monte Carlo runs	Average over 100 independent runs per point (Figs. 6, 8, 9); single-run trajectory (Fig. 7)

- **Chen's algorithm [17]:** This algorithm mainly introduces the early frame termination mechanism in DFSA for RFID. The basic device estimation is obtained by multiplying the number of collided slots in a frame by 2.39. When 20% of the frame has been completed, the reader checks whether an early break should be triggered by comparing the gap from the prior estimation to the posterior estimation obtained in the first 20% of the frame. If triggered, the reader immediately initiates a new frame according to the posterior estimation. If not triggered, the reader completes the current frame and initiates a new frame with the same configuration as the previous one.

Fig. 6 presents the RA efficiency and TST performance of the three algorithms as the device population size varies, where the performance for each population size is the average of 100 repeated simulation results. Note that the reader does not know the device population size initially in the simulation. The results show that the RA efficiencies of both the proposed BE^{2}FT and Chen's algorithm remain nearly constant as the device population size increases. The stability of the proposed BE^{2}FT arises from its ability to estimate the number of devices awaiting UID reporting in real time and adaptively configure the frame length, i.e., the number of Msg1 response opportunities L. Compared with the maximum RA efficiencies shown in Fig. 4(b), which are obtained under the assumption that the number of devices is perfectly known at the beginning of each frame, BE^{2}FT achieves nearly identical performance. This confirms both the accuracy of its estimation mechanism and the effectiveness of its adaptive control of L. For comparison, we also plot an RA efficiency curve with a fixed L across different device population sizes, where the RA efficiency fluctuates significantly as N varies.When it comes to TST, it increases proportionally with the device population size for both BE^{2}FT and Chen's algorithm. The proposed BE^{2}FT outperforms Chen's algorithm and achieves performance close to that of the ideal algorithm.

B. Performance Comparison in Multi-Channel A-IoT System

We return to the primary focus of the multi-channel A-IoT system. In the following comparisons, in addition to BEst, BE^{2}FT, and the ideal algorithm, we also introduce the following system:

- **FDMA RFID [22]:** This frame structure also considers multi-channel RA that only considers the UID response without sending RID. Each Query (like paging in A-IoT for the first slot) or QueryRep (like slot trigger in A-IoT for the subsequent slots) in each slot triggers only one Msg3 response opportunity. Additionally, the simulations assume that the reader is aware of the number of devices and determines the number of slots in a frame accordingly.

We examine the effectiveness of the estimation capability of the proposed BEst and BE^{2}FT algorithms in an A-IoT system employing B-RAP with $K = 4$. Fig. 7 illustrates the number of devices awaiting UID reporting in the system and their estimated values over time, where 10,000 devices initially need to report their UIDs to the reader. It is worth noting that this scenario also represents a bursty traffic condition, as it can be interpreted as 10,000 devices suddenly attempting to access the network simultaneously. Each algorithm is evaluated over 50 independent simulation runs to obtain statistically reliable results. Since the reader does not initially know the number of devices in the system, it sets the initial estimate to 100. Consequently, the reader must adjust its estimation in the early phase to optimize system operation. BE^{2}FT clearly tracks the number of devices much faster, whereas BEst converges considerably more slowly. The superior performance of BE^{2}FT arises from its early frame termination mechanism, which enables rapid adjustment of L. In Fig. 7(b), the box symbols indicate the time instants when early frame termination occurs. It can be observed that these events appear more frequently at the beginning and end of the inventory process, while in the middle phase the algorithm operates with appropriate frame configurations without triggering termination.

We evaluate the aforementioned control algorithms in A-IoT systems with $K = 4$. Fig. 8 presents the RA efficiency

Authorized licensed use limited to: Higher School of Economics. Downloaded on April 07,2026 at 10:42:35 UTC from IEEE Xplore. Restrictions apply.

and TST performance as the device population size varies, where each data point represents the average of 100 simulation runs. The results show that the RA efficiency remains nearly constant across different device population sizes, and the efficiencies of all the algorithms are close to their corresponding maximum values observed in Fig. 4(b). The fluctuating RA efficiency curve for the fixed $L = 200$ case further confirms the effectiveness of the BEst and BE^{2}FT algorithms in estimating the number of devices online and adaptively controlling the L value. For the ideal algorithm, which assumes the number of devices is perfectly known at any time, E-RAP achieves approximately a 37% improvement in RA efficiency compared to B-RAP, while MRMC-RAP yields an increase of around 53%. The performance of the FDMA RFID scheme lies between those of B-RAP and E-RAP. The TST results also corroborate these trends. Among the algorithms for each RAP scheme, we observe that the proposed BE^{2}FT algorithm achieves performance that is already very close to that of the ideal algorithm and is also robust, in terms of both RA efficiency and TST. In contrast, BEst still exhibits a noticeable gap compared with the ideal algorithm.

C. Capture Effect

We investigate the effect of the capture phenomenon in A-IoT by evaluating the RA efficiency for $K = 4$ and a device population of 3000, as shown in Fig. 9. The capture probability p_{ca} varies from 0 to 0.5. The ideal algorithm is assumed to know p_{ca} and adjusts L according to (30), while BE^{2}FT and BEst operate without this information. As p_{ca} increases, more Msg1 responses are successfully captured from collisions, leading to improved RA efficiency. Notably, the RA efficiency of E-RAP converges toward that of MRMC-RAP as p_{ca} increases. According to (29), when p_{ca} increases to 0.5, the expected number of successful Msg1 responses per opportunity increases to approximately $0.5K$, implying that MRMC-RAP effectively requires two Msg1 opportunities per slot, which aligns with the slot structure of E-RAP. The proposed BE^{2}FT achieves performance close to the ideal algorithm, mainly because its early frame termination mechanism is effectively triggered to compensate for inaccuracies resulting from the capture effect, which helps prevent the use of suboptimal frame lengths.

VII. Conclusion

In this paper, we presented a comprehensive study on RA for multi-channel A-IoT networks, delivering the first analytically grounded RAP frameworks for A-IoT within 3GPP Release 19. Building on the FDMA capabilities supported for A-IoT, in addition to the B-RAP and E-RAP introduced by 3GPP, we proposed the MRMC-RAP scheme. Closed-form expressions for RA efficiency were derived under each RAP scheme, and optimal access configurations were analytically and numerically characterized. Furthermore, to account for practical uncertainty in device population size, we developed an estimation algorithm, BE^{2}FT, that enables adaptive access control even under bursty traffic conditions. Simulation results highlighted the advantages: the proposed MRMC-RAP consistently achieved the best RA efficiency across scenarios, and BE^{2}FT delivered performance close to the ideal case where the number of devices is perfectly known at any time and access parameters are optimally configured.Tailored for 6G A-IoT, the proposed RA designs support reader-driven large-scale inventory operations and can be extended to broader A-IoT use cases, such as asset tracking, smart retail, industrial monitoring, and urban sensing, under massive device densities.

References

[1] M. M. Butt et al., "Ambient IoT: A missing link in 3GPP IoT devices landscape," *IEEE Internet Things Mag.*, vol. 7, no. 2, pp. 85–92, Mar. 2024.

[2] M. Shirvanimoghaddam et al., "Towards a green and self-powered Internet of Things using piezoelectric energy harvesting," *IEEE Access*, vol. 7, pp. 94533–94556, 2019.

[3] C. Xu, L. Yang, and P. Zhang, "Practical backscatter communication systems for battery-free Internet of Things: A tutorial and survey of recent research," *IEEE Signal Process. Mag.*, vol. 35, no. 5, pp. 16–27, Sep. 2018.

[4] K. Zheng, R. Xu, J. Mei, H. Yang, L. Lei, and X. Wang, "Ambient IoT toward 6G: Standardization, potentials, and challenges," *IEEE Access*, vol. 12, pp. 146668–146677, 2024.

[5] *Study on Solutions for Ambient IoT (Internet of Things) in NR*, Standard (TR) 38.769 (V19.0.0), Release 19, 3GPP, Dec. 2024.

[6] *New Work Item: Solutions for Ambient IoT (Internet of Things) in NR*, Standard RP-243326, 3GPP, Dec. 2024.

[7] *New WID on Solutions for Ambient IoT (Internet of Things) in NR Phase 2*, Standard RP-251841, 3GPP, Jun. 2025.

[8] D. C. Nguyen et al., "6G Internet of Things: A comprehensive survey," *IEEE Internet Things J.*, vol. 9, no. 1, pp. 359–383, Jan. 2022.

[9] *Study on Ambient Power-Enabled Internet of Things*, Standard (TR) 22.840 (V19.0.0), Release 19, 3GPP, Dec. 2023.

[10] *EPC Radio-Frequency Identity Generation-2 UHF RFID Standard: Specification for RFID Air Interface Protocol for Communications at 860 MHz 930 MHz, Version 3.0*, EPCglobal Inc., Saint Charles, MO, USA, Jan. 2024.

[11] A. N. Arun et al., "Ambient IoT: Communications enabling precision agriculture," *IEEE Commun. Mag.*, vol. 63, no. 4, pp. 137–143, Apr. 2025.

[12] M. Li, Y. Huang, F. R. Yu, P. Si, and H. Zhang, "Ambient backscatter communication-assisted intelligent resource management for green industrial IoT," *IEEE Wireless Commun.*, vol. 32, no. 1, pp. 174–181, Feb. 2025.

[13] R. Singh, A. K. Yerrapragada, and R. K. Ganti, "Physical layer design for ambient IoT," *IEEE Commun. Standards Mag.*, vol. 9, no. 4, pp. 224–231, Dec. 2025.

[14] S. Malik, J. Saini, D. N. Amudala, and R. Budhiraja, "Coverage analysis and enhancement techniques for ambient IoT technology," *IEEE Commun. Standards Mag.*, vol. 9, no. 4, pp. 232–239, Dec. 2025.

[15] H. Vogt, "Multiple object identification with passive RFID tags," in *Proc. IEEE Int. Conf. Syst., Man Cybern.*, vol, vol. 3, Feb. 2002, p. 6.

[16] W.-T. Chen, "An accurate tag estimate method for improving the performance of an RFID anticollision algorithm based on dynamic frame length ALOHA," *IEEE Trans. Autom. Sci. Eng.*, vol. 6, no. 1, pp. 9–15, Jan. 2009.

[17] W.-T. Chen, "Optimal frame length analysis and an efficient anti-collision algorithm with early adjustment of frame length for RFID systems," *IEEE Trans. Veh. Technol.*, vol. 65, no. 5, pp. 3342–3348, May 2016.

[18] H. Salah and G. Gaydadjiev, "Toward optimal RFID frame lengths," *IEEE J. Radio Freq. Identificat.*, vol. 7, pp. 83–90, 2023.

[19] F. Schoute, "Dynamic frame length ALOHA," *IEEE Trans. Commun.*, vol. COM-31, no. 4, pp. 565–568, Apr. 1983.

[20] H. Vogt, "Efficient object identification with passive RFID tags," in *Pervasive Computing*. Berlin, Germany: Springer, Jan. 2002, pp. 98–113.

[21] J. Su, Z. Sheng, D. Hong, and G. Wen, "An effective frame breaking policy for dynamic framed slotted ALOHA in RFID," *IEEE Commun. Lett.*, vol. 20, no. 4, pp. 692–695, Apr. 2016.

[22] Z. Huang et al., "Frequency division multiple access extension of standard UHF RFID systems for multiple tags inventory with successive interference cancellation," *IEEE Internet Things J.*, vol. 12, no. 12, pp. 19615–19630, Jun. 2025.

Authorized licensed use limited to: Higher School of Economics. Downloaded on April 07,2026 at 10:42:35 UTC from IEEE Xplore. Restrictions apply.

[23] H.-H. Choi, W. Shin, M. Levorato, and H. V. Poor, "Harvest-or-access: Slotted ALOHA for wireless powered communication networks," *IEEE Trans. Veh. Technol.*, vol. 68, no. 11, pp. 11394–11398, Nov. 2019.
[24] M. Ghanbarinejad and C. Schlegel, "Irregular repetition slotted ALOHA with multiuser detection," in *Proc. 10th Annu. Conf. Wireless On-Demand Netw. Syst. Services (WONS)*, Mar. 2013, pp. 201–205.
[25] E. Paolini, G. Liva, and M. Chiani, "Coded slotted ALOHA: A graph-based method for uncoordinated multiple access," *IEEE Trans. Inf. Theory*, vol. 61, no. 12, pp. 6815–6832, Dec. 2015.
[26] U. Demirhan and T. M. Duman, "Irregular repetition slotted ALOHA with energy harvesting nodes," *IEEE Trans. Wireless Commun.*, vol. 18, no. 9, pp. 4505–4517, Sep. 2019.
[27] J. Haghighat and T. M. Duman, "Analysis of coded slotted ALOHA with energy harvesting nodes for perfect and imperfect packet recovery scenarios," *IEEE Trans. Wireless Commun.*, vol. 22, no. 11, pp. 7424–7437, Nov. 2023.
[28] H. Jin, W. T. Toor, B. C. Jung, and J.-B. Seo, "Recursive Pseudo–Bayesian access class barring for M2M communications in LTE systems," *IEEE Trans. Veh. Technol.*, vol. 66, no. 9, pp. 8595–8599, Sep. 2017.
[29] N. Jiang, Y. Deng, and A. Nallanathan, "Traffic prediction and random access control optimization: Learning and non-learning-based approaches," *IEEE Commun. Mag.*, vol. 59, no. 3, pp. 16–22, Mar. 2021.
[30] J. Liu, M. Agiwal, M. Qu, and H. Jin, "Online control of preamble groups with priority in massive IoT networks," *IEEE J. Sel. Areas Commun.*, vol. 39, no. 3, pp. 700–713, Mar. 2021.
[31] H. S. Jang, H. Jin, B. C. Jung, and T. Q. S. Quek, "Versatile access control for massive IoT: Throughput, latency, and energy efficiency," *IEEE Trans. Mobile Comput.*, vol. 19, no. 8, pp. 1984–1997, Aug. 2020.
[32] S. K. Nobar, M. H. Ahmed, Y. Morgan, and S. A. Mahmoud, "Performance analysis of LTE random access protocol with an energy harvesting M2M scenario," *IEEE Internet Things J.*, vol. 7, no. 2, pp. 893–905, Feb. 2020.
[33] H. Xie, W. T. Toor, J.-B. Seo, and H. Jin, "Dual-core: Dual collision resolution for massive access in cellular IoT networks," *IEEE Trans. Commun.*, vol. 73, no. 9, pp. 7398–7414, Sep. 2025.
[34] W. Fan, P. Fan, and Y. Long, "Joint delay-energy optimization for multi-priority random access in machine-type communications," *IEEE Trans. Wireless Commun.*, vol. 23, no. 2, pp. 1416–1431, Feb. 2024.
[35] *Final Report of 3GPP TSG RAN WG1 #121 v1.0.0*, Standard R1-2505102, 3GPP, 3GPP TSG RAN WG1 Meeting #121, St Julian's, Malta, May 2025.
[36] J. Su, Z. Sheng, A. X. Liu, Y. Han, and Y. Chen, "Capture-aware identification of mobile RFID tags with unreliable channels," *IEEE Trans. Mobile Comput.*, vol. 21, no. 4, pp. 1182–1195, Apr. 2022.



Shilun Song (Member, IEEE) received the B.S. degree in mathematics and applied mathematics from Harbin Institute of Technology (HIT), Weihai, China, in 2019. He is currently pursuing the combined M.S. and Ph.D. degree with the Department of Electrical and Electronic Engineering, Hanyang University, Ansan, South Korea. His research interests include 6G communications, random access networks, and communication systems aligned with 3GPP standardization.



Huan Zhang received the B.S. degree in electromagnetic field and wireless technology from Harbin Institute of Technology (HIT), Weihai, China, in 2022. She is currently pursuing the combined M.S. and Ph.D. degree with the Department of Electrical and Electronic Engineering, Hanyang University, Ansan, South Korea. Her research interests include 6G communications, quantum computing, and quantum communication. She was a recipient of the Best Paper Award of the International Symposium on Satellite Communication Systems and Services in 2024.



Huiyang Xie received the B.S. degree in information and computing science from Harbin Institute of Technology (HIT), Weihai, China, in 2020. She is currently pursuing the combined M.S. and Ph.D. degree with the Department of Electrical and Electronic Engineering, Hanyang University, Ansan, South Korea. Her research interests include the Internet of Things, with a particular focus on timely status update (age of information) and random access networks. She was a recipient of the Best Workshop Paper Award of International Conference on ICT Convergence in 2023.



Hu Jin (Senior Member, IEEE) received the B.E. degree in electronic engineering and information science from the University of Science and Technology of China, Hefei, China, in 2004, and the M.S. and Ph.D. degrees in electrical engineering from Korea Advanced Institute of Science and Technology, Daejeon, South Korea, in 2006 and 2011, respectively. From 2011 to 2013, he was a Post-Doctoral Fellow with The University of British Columbia, Vancouver, BC, Canada. From 2013 to 2014, he was a Research Professor with Gyeongsang National University, Tongyeong, South Korea. Since 2014, he has been with the School of Electrical Engineering, Hanyang University, Ansan, South Korea, where he is currently a Professor. His research interests include medium-access control and radio resource management for random access networks and scheduling systems considering advanced signal processing and queueing performance.

Authorized licensed use limited to: Higher School of Economics. Downloaded on April 07,2026 at 10:42:35 UTC from IEEE Xplore. Restrictions apply.